

SULIT



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II 2021/2022

SUBJECT CODE : SECR2043
SUBJECT NAME : OPERATING SYSTEM
SECTION : SECR/J/B/V/P/H/ SCSR/J/B/V/P/H
TIME : PAPER 2 – 90 MINUTES
DATE/DAY : 18 JULY 2022 (MONDAY)
VENUES : Online

INSTRUCTIONS:

Answer all questions from Paper 1 and Paper 2. Show all your works.

This test will contribute 35% towards the total marks of 100%.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	TEH JING LING
I/C No.	010111-08-0090
Year / Course	2/SECJH
Section (Circle)	01 / 02 / 03 / 04 / 05 / 06 / 07 / 08 / 09 / 15
Lecturer Name (Circle)	Dr Zuriahadi / Mrs Rashidah Dr Nur Haliza / Dr Nor Shahida

This question paper consists of 4 printed pages excluding this page.

Question 3

a) i) root / programs / p / F6

root / bin / recorder / recorder / first / F19

ii) mail / F7

all / F7

iii) cd /spell/mail/prt

iv) Relative filename

v)	Owner	Group	public
	6	5	1
	110	101	001
	RW-	R-D	--D

Owner allowed only read and write File F16.

Group allowed only read and delete File F16.

Public allowed only delete File F16.

vi) -rwxr--wx 7 research admin 512 Sep 3 08:30 F16

vii) "public.doc" is a file owned by "research" with group "admin".

"public/" is a subdirectory owned by "research" with group "librarian".

viii) chmod 444 *.doc

chmod a+r,a-wx *.doc.

Question 4

a)

	4	3	2	1	2	3	4	7	3	2	1	2	4	5	1	2	1	3	2	1
Frame #0	4	4	4	4				7					7	7		7	1	1		
Frame #1		3	3	3				3					4	4		4	4	3		
Frame #2			2	2				2					2	5		5	5	5		
Frame #3				1				1					1	1		2	2	2		

No. of page fault = 10

b)

	4	3	2	1	2	3	4	7	3	2	1	2	4	5	1	2	1	3	2	1
Frame #0	4	4	4	1				1	1	3	3	3	4	4	4	1	1			
Frame #1		3	3	3				4	4	4	2	2	2	5	5	5	3			
Frame #2			2	2				2	7	7	7	1	1	1	2	2	2			

No. of page fault = 14

c) 4 frames have less 4 page faults than 3 frames by using FIFO. $(14 - 10 = 4)$

More frame number can reduce page fault amount.

The system efficiency at 4 frames is greater than system efficiency of 3 frames. Since the higher the frame number, the lower the page fault amount.

Belady's Anomaly

More frames may lead to more page faults.

d)

	4	3	2	1	2	3	4	7	3	2	1	2	4	5	1	2	1	3	2	1
Frame #0	4	4	4	4				7					4	5						
Frame #1		3	3	3				3					3	3						
Frame #2			2	2				2					2	2						
Frame #3				1				1					1	1						

No. of page fault = 7

e) OPT have less 3 page faults than FIFO $(10 - 7 = 3)$

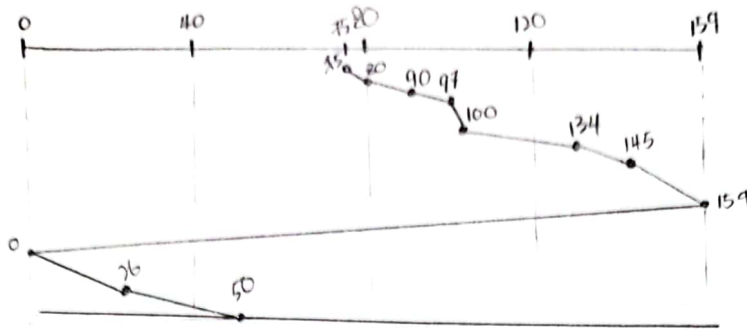
The system efficiency of OPT is greater than the system efficiency of FIFO since it has less page fault number compared to FIFO.

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Question 5

5 6 2 7 1 3 11
97, 100, 50, 134, 26, 145, 80, 90
26, 50, 80, 90, 97, 100, 134, 145

a)



b) Total distance
 $= 159 - 75 + 159 - 0 + 50 - 0$
 $= 84 + 159 + 50$
 $= 293 \text{ cylinders.}$

c) Total distance
 $= 159 - 75 + 159 - 26$
 $= 217 \text{ cylinders.}$

Difference
 $= 293 \text{ cylinders} - 217 \text{ cylinders} = 76 \text{ cylinders.}$

The total distance of SCAN is less than C-SCAN about 76 cylinders.

SCAN is faster than C-SCAN.

The disk arm starts at one end of the disk, and moves toward the other end, servicing requests until it gets to the other end of the disk, where the head movement is reversed and servicing continues.

d)

